Hamdard University
Department Of Computing

FYP Primilinary Report

Web-Based Phishing Detection System (FYP-017/FA25)

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In partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science Engineering 2025

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<u>Certificate of Approval</u>

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This project "<u>Web-Based Phishing Detection System</u>" is presented by <u>M. Ibtehaj Mubarak, M. Zaid Bin Saud, and M. Azaan Rashid</u> under the supervision of their project advisor and approved by the project examination committee, and acknowledged by the Hamdard Institute of Engineering and Technology, in the fulfillment of the requirements for the Bachelor degree of <u>Computer Science Engineering</u>.

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Authors' Declaration

We declare that this project report was carried out in accordance with the rules and regulations of Hamdard University.

The work is original except were indicated by special references in the text and no part of the report has been submitted for any other degree. The report has not been presented to any other University for examination.

Dated: 27/07/2025

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Plagiarism Undertaking

We, M. Ibtehaj Mubarak, M. Zaid Bin Saud, and M. Azaan Rashid, solemnly declare that the work presented in the Final Year Project Report titled “Web-Based Phishing Detection System” has been carried out solely by ourselves with no significant help from any other person except few of those which are duly acknowledged.

We confirm that no portion of our report has been plagiarized and any material used in the report from other sources is properly referenced.

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Definition of Terms, Acronyms, and Abbreviations

Term	Description
Phishing	A cyberattack where attackers trick users into revealing sensitive information using fake links or websites.
Phishing URL	A web link designed to redirect users to fake or malicious websites.
Rule-Based Detection	A detection method that uses predefined rules (e.g., URL length, special characters, IP address usage) to identify phishing links.
Blacklist	A database containing known malicious or phishing URLs used to block unsafe websites.
Random Forest	An ensemble learning method for classification, regression, and other tasks that operates by constructing multiple decision trees
Web-Application	A software application that runs on a web browser without requiring installation.
Machine Learning	A trained algorithm that learns patterns from data to make predictions or classifications.
Random Forest Classifier	A supervised machine learning algorithm that uses multiple decision trees to classify URLs as phishing or legitimate.
AI Chatbot	A rule-based or AI-powered interface that interacts with users and provides guidance or responses.

Abstract

Phishing attacks have become one of the most common and harmful cybersecurity threats, targeting users through deceptive links distributed via emails, social media platforms, and messaging applications. These attacks often result in data theft, financial fraud, and unauthorized access to sensitive information. Due to a lack of technical awareness and real-time detection tools, many users unknowingly fall victim to such malicious activities, especially while browsing the web.

This project proposes a **Web-Based Phishing Detection System** designed to detect, block, and alert users about suspicious or phishing URLs in real time. The system combines rule-based detection techniques, blacklist API integration, and machine learning models to analyze URL structures and identify potential threats. The rule-based engine examines features such as URL length, presence of IP addresses, special characters, suspicious keywords, and HTTPS availability, while blacklist services verify URLs against known phishing databases.

To enhance detection accuracy, a Machine Learning classifier (Random Forest) is integrated into the backend to predict phishing links based on extracted URL features. This hybrid approach improves reliability by handling both known phishing patterns and previously unseen threats. The system is implemented as a web application using HTML, CSS, and JavaScript for the frontend and Python (Flask) for backend processing, ensuring lightweight performance and ease of access across devices.

Additionally, the system includes a chat-based assistance module that provides users with clear warnings, explanations, and guidance based on detection results. The proposed solution aims to improve user awareness, reduce phishing-related risks, and provide a scalable foundation for future AI-based cybersecurity enhancements.

Keywords:

Phishing Detection, Web-Based Security System, Rule-Based Analysis, Machine Learning, Random Forest Classifier, Blacklist APIs, URL Analysis, Cybersecurity, Phishing Prevention, Chatbot Assistance

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Chapter 1: Introduction

1.1 Background

With the rapid growth of internet usage, phishing attacks have emerged as a major cybersecurity threat worldwide. Phishing is a fraudulent practice in which attackers deceive users by creating fake websites or malicious links that appear legitimate. These attacks are commonly carried out through emails, social media platforms, and messaging applications, targeting users to steal sensitive information such as login credentials, banking details, or personal data.

Despite advancements in cybersecurity technologies, phishing attacks continue to increase due to their simplicity and effectiveness. Many users lack the technical knowledge required to identify malicious links before clicking on them. Existing security solutions are often complex, resource-heavy, or focused on enterprise-level environments rather than general web users. This creates a strong need for a lightweight, accessible, and real-time phishing detection solution that can assist users during everyday web usage.

1.2 Problem Statement

Phishing attacks have become one of the most common and dangerous cybersecurity threats on the internet. Attackers create fake or malicious websites and links that appear legitimate in order to trick users into revealing sensitive information such as usernames, passwords, banking details, or personal data.

With the increasing use of online services, users frequently receive suspicious links through emails, social media platforms, and messaging applications.Most internet users lack the technical knowledge and tools required to identify phishing links before clicking on them.

Existing security solutions are often complex, require installation, or are focused on enterprise-level systems rather than general users. Additionally, many lightweight systems rely only on static blacklist checks, which are ineffective against newly generated or modified phishing URLs.

There is a need for a simple, web-based, and real-time phishing detection system that can analyze URLs efficiently without requiring advanced technical expertise from users. Such a system should combine rule-based analysis, blacklist verification, and Machine Learning techniques to improve detection accuracy while remaining lightweight and user-friendly.

This project aims to address these challenges by developing this project that detects, blocks, and alerts users about phishing links in real time, while providing clear guidance and maintaining scalability for future security enhancements.

1.3 Objectives of the Project

The main objectives of this project are:

- To design and develop a web-based system for detecting phishing URLs
- To analyze URLs using rule-based logic and blacklist APIs
- To integrate a machine learning model for improved phishing prediction
- To block and warn users about malicious or suspicious links
- To provide user guidance through a chatbot-based assistance module
- To create a lightweight, user-friendly, and scalable security solution

1.4 Scope of the Project

The scope of this project is limited to the development of a web-based phishing detection system that analyzes user-provided URLs. The system focuses on detecting phishing attempts using predefined rules, blacklist services, and a machine learning classifier. The application provides real-time feedback and guidance to users through a web interface.

The project does not aim to replace full-scale antivirus solutions but serves as an awareness and protection tool for everyday web users.

Future enhancements may include automatic URL detection, browser extensions, and advanced AI-based threat analysis.

1.5 Methodology Overview

The project follows the **Waterfall Software Development Model**, which ensures a structured and sequential development process. Each phase like requirements analysis, system design, implementation, testing, and documentation – is completed before moving to the next.

This approach is suitable for academic projects with clearly defined objectives and deliverables.

The phishing detection mechanism uses a hybrid approach, combining rule-based analysis, blacklist verification, and a machine learning model (Random Forest Classifier) to improve detection accuracy and system reliability.

1.6 Significance of the Project

This project contributes to cybersecurity awareness by providing users with a simple and effective tool to detect phishing threats. It helps reduce the risk of data theft and financial fraud by preventing users from accessing malicious links.

Additionally, the project demonstrates the practical application of rule-based systems and machine learning in real-world cybersecurity problems, making it valuable for both academic and practical use.

1.7 Organization of the Report

This report is organized into multiple chapters. Chapter 1 introduces the project, its objectives, scope, and significance.

Chapter 2 provides an overall description of the system. Chapter 3 discusses the external interface requirements. Subsequent chapters cover system design, implementation details, testing, results, and conclusions.

Chapter 2: Overall Description

2.1 Product Perspective

The Web-Based Phishing Detection System is a standalone web application designed to help users identify and avoid phishing attacks by analyzing suspicious URLs. The system does not replace existing antivirus or browser security tools; instead, it acts as an additional protective and awareness layer for users.

The application follows a client–server architecture. The frontend provides a simple interface where users can enter URLs, while the backend processes the input using rule-based logic, blacklist APIs, and a machine learning model. The system is modular in design, allowing future enhancements such as browser extensions, automatic URL detection, and advanced AI-based analysis.

2.2 Product Functions

The major functions of the system include:

- Accepting URLs entered by users through a web interface
- Analyzing URLs using rule-based detection techniques
- Verifying URLs against known phishing databases through blacklist APIs
- Predicting phishing likelihood using a ML model (Random Forest Classifier)
- Displaying real-time results such as Safe, Suspicious, or Phishing
- Blocking or warning users about malicious links
- Providing guidance and awareness messages through a chatbot interface

2.3 User Classes and Characteristics

The system is designed for the following user classes:

- **General Internet Users:**
  Users with limited cybersecurity knowledge who want a simple way to verify suspicious links.

- **Students and Academic Users:**
  Individuals using the system for learning and awareness about phishing attacks.

- **Non-Technical Users:**
  Users who require a straightforward interface without technical complexity.

2.4 Operating Environment

The system will operate in the following environment:

- **Frontend:**
  Web browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge

- **Backend:**
  Python-based server using Flask framework

- **Machine Learning Environment:**
  Scikit-learn library for model training and prediction

- **Platform:** Any device with an internet connection or web browser

2.5 Design and Implementation Constraints

The following constraints apply to the system:

- Dependence on third-party blacklist APIs, which may have usage limits
- Machine learning model accuracy depends on dataset quality
- System requires an active internet connection for API-based checks
- Development is limited to the tools and time available within FYP duration

2.6 User Documentation

The system will be supported with the following documentation:

- User guide explaining how to scan URLs
- System overview for understanding detection results
- Project documentation including project plan, design documents, and reports

2.7 Assumptions and Dependencies

The system is developed based on the following assumptions:

- Users will manually input URLs for scanning
- Blacklist APIs provide updated and reliable phishing data
- The ML model is trained on a representative phishing dataset
- The system will be used for awareness and preventive purposes

Chapter 3: Literature Review & Related Work

3.1 Introduction

Phishing attacks are one of the most prevalent cybersecurity threats on the internet, targeting users through deceptive websites and malicious links. Over the years, researchers have proposed various techniques to detect and prevent phishing attacks, including rule-based systems, blacklist approaches, heuristic analysis, and machine learning models.

This chapter reviews existing research work related to phishing detection and highlights the strengths and limitations of different approaches. The reviewed literature helps justify the selection of techniques used in our project.

3.2 Rule-Based Phishing Detection Approaches

Rule-based phishing detection systems analyze URLs or webpages using predefined rules that represent common phishing characteristics. Basnet et al. (2015) proposed a rule-based method that examines attacker tactics such as abnormal URL structures, suspicious redirections, and misleading domain names. Their system achieved an accuracy between 95% and 99%, with a low false positive rate of 0.5% to 1.5%.

These results demonstrate that carefully designed rule-based systems can be highly effective, especially for detecting known phishing patterns. Such approaches are lightweight and suitable for real-time applications, making them ideal for web-based and user-focused security systems.

3.3 Heuristic and Blacklist-Based Detection Systems

Heuristic and blacklist-based techniques are widely used in phishing detection. Researchers at Temple University introduced *MobiFish*, a mobile anti-phishing system that relies on heuristic rules combined with blacklist verification. Their study showed that while blacklist methods are effective for known phishing sites, they struggle to detect zero-day phishing attacks.

Despite this limitation, the combination of heuristics and blacklist APIs remains a practical and efficient solution for detecting common phishing threats. This approach is particularly suitable for lightweight systems where performance and response time are critical.

3.4 Machine Learning-Based Phishing Detection

Machine learning techniques have gained popularity in phishing detection due to their ability to identify complex patterns in data. Various classifiers such as Logistic Regression, Decision Trees, Random Forest, and Naive Bayes have been used to predict phishing URLs based on extracted features like URL length, special characters, and suspicious keywords.

Random Forest classifiers, in particular, have shown strong performance due to their robustness and ability to handle feature variability. However, machine learning models often require large datasets and computational resources, which may not always be practical for lightweight applications.

3.5 Hybrid Detection Approaches

Recent studies emphasize the effectiveness of hybrid phishing detection systems that combine rule-based logic, blacklist checks, and machine learning models. Dholakia and Agrawal (2018) conducted a comprehensive review of phishing detection techniques and concluded that no single method can handle all phishing threats effectively. Instead, a combination of approaches provides better accuracy and adaptability.

Hybrid systems balance detection accuracy with system efficiency, making them suitable for real-time web-based security applications.

3.6 Summary

This chapter reviewed existing phishing detection techniques, including rule-based systems, blacklist approaches, machine learning models, and hybrid solutions.

The literature indicates that rule-based and blacklist methods are efficient for detecting known phishing attacks, while machine learning improves detection of complex and unseen threats.

Based on these findings, the proposed system adopts a hybrid approach to achieve accurate, lightweight, and scalable phishing detection.